

**A NOVACARE INSURANCE
COMPARATIVE ASSESSMENT OF
THREE MODEL CONFIGURATIONS
TESTED DURING DEVELOPMENT —
WITH FINDINGS, JUSTIFICATION FOR
THE FINAL MODEL SELECTION, AND
RECOMMENDATIONS.**

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1. Executive Summary

Three model configurations were evaluated during this project. The final model — a Logistic Regression trained on the full 6,150-claim dataset with icd10_description included — was selected for the Power BI dashboard and client submission. This report documents all three configurations, compares their performance, and justifies the final selection.

Config	Dataset	ICD Description	Status
Model A	Full dataset (6,150 claims)	With icd10_description	FINAL — submitted
Model B	Full dataset (6,150 claims)	Without icd10_description	Comparison only
Model C	No duplicates (5,670 claims)	With icd10_description	Comparison only

2. Comparative Model Performance

The table below presents all key performance metrics side by side for the three configurations tested. All models were trained using Logistic Regression with class_weight='balanced', random_state=42, and an 80/20 stratified train-test split.

Metric	Model A (Final — With ICD)	Model B (No ICD)	Model C (No Duplicates)
Total Claims	6,150	6,150	5,670
Denial Rate	39.58%	39.58%	35.15%
Test Set Size	1,230	1,230	1,134
Model Accuracy	63.98%	63.41%	50.53%
Recall — Denied	28.13%	30.39%	48.12%
Precision — Denied	59.57%	57.14%	35.16%
F1 Score	0.3821	0.3968	0.4063
ROC-AUC	0.5760	0.5761	0.4982
True Negatives (TN)	650	632	N/A
False Positives (FP)	93	111	N/A
False Negatives (FN)	350	339	N/A
True Positives (TP)	137	148	N/A
High-Risk Flagged	1,237	1,356	2,799
Value at Risk	\$3.68M	\$4.10M	\$9.09M
FN Leakage Estimate	\$1.04M	\$1.01M	N/A

3. Metric-by-Metric Analysis

3.1 Accuracy

Model A achieved the highest accuracy at 63.98%, marginally ahead of Model B (63.41%). Model C scored only 50.53% — barely better than random guessing. However, accuracy is a limited metric for imbalanced classification problems. A model that predicted 'not denied' for every single claim in Model A's dataset would still achieve 60.42% accuracy. Accuracy must always be read alongside recall and precision.

3.2 Recall — The Priority Metric

Recall measures what percentage of actual denials the model successfully caught. For NovaCare, every missed denial (False Negative) represents direct revenue leakage — a claim submitted without a risk flag that subsequently gets rejected. This makes recall the single most important metric in this business context.

Model C achieved the highest recall at 48.12%, catching nearly half of all actual denials in the no-duplicates dataset. Model B achieved 30.39% and Model A (final) achieved 28.13%. The lower recall in Models A and B is explained by the presence of `is_duplicate_claim` as a dominant feature: the model concentrates its predictive power on the easily-identifiable duplicate claims and is consequently less sensitive to the subtler patterns driving non-duplicate denials.

3.3 Precision

Precision measures how reliable the high-risk flags are — of all claims flagged as likely to be denied, what percentage actually were denied? Model A achieved the highest precision at 59.57%, meaning roughly 6 in 10 flagged claims were genuinely at risk. Model B achieved 57.14% and Model C a much lower 35.16%. Low precision in Model C means the billing team would receive a large volume of false alarms (2,799 flagged vs 1,237 in Model A), creating unnecessary workload.

3.4 F1 Score

The F1 score balances recall and precision into a single number. Model C achieves the highest F1 (0.4063) because its recall is substantially higher, and recall is weighted equally with precision in the F1 formula. Model B (0.3968) marginally outperforms Model A (0.3821). However, F1 does not account for the cost asymmetry between False Positives (minor extra review work) and False Negatives (real revenue loss), making it a useful but incomplete guide for final model selection.

3.5 ROC-AUC

ROC-AUC measures the model's ability to rank denied claims above non-denied claims. Models A and B are nearly identical at 0.5760 and 0.5761 respectively — both marginally above random (0.5), indicating modest but consistent discrimination ability. Model C drops to 0.4982, effectively random, confirming that removing duplicate claims eliminates the model's clearest separation signal.

3.6 Confusion Matrix Comparison

Model A (Final — With ICD Description):

	Predicted: Not Denied	Predicted: Denied
Actual: Not Denied	True Negative 650 Correctly cleared	False Positive 93 Extra review

Actual: Denied	False Negative 350 Missed denial FINANCIAL RISK	True Positive 137 Correctly flagged
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Model B (Without ICD Description):

	Predicted: Not Denied	Predicted: Denied
Actual: Not Denied	True Negative 632 Correctly cleared	False Positive 111 Extra review
Actual: Denied	False Negative 339 Missed denial FINANCIAL RISK	True Positive 148 Correctly flagged

Model A produces 18 fewer false positives than Model B (93 vs 111), meaning the billing team receives fewer unnecessary flags. Model B catches 11 more true positives (148 vs 137) but at the cost of 18 more false alarms. Depending on the billing team's capacity, Model A's precision-oriented profile may be more operationally practical.

4. Why Model A (With ICD Description) Was Selected

The final model submitted for the Power BI dashboard and client presentation is Model A — the full 6,150-claim dataset with icd10_description included. The decision rests on four grounds:

Highest Accuracy (63.98%)

Model A correctly classifies the largest percentage of all claims, providing the most reliable overall picture for the CFO dashboard.

Highest Precision (59.57%)

Precision directly affects the billing team's workload. Model A generates the fewest false alarms (93 vs 111 in Model B and 2,799 in Model C), making its flags the most actionable.

Full Dataset Integrity

Model C removes 480 duplicate claims from training. While this reveals subtler denial patterns, it also removes a real and operationally significant denial driver. The full dataset model reflects NovaCare's actual claims environment more faithfully.

Stable AUC with Improved Precision

Model A's AUC (0.5760) is virtually identical to Model B (0.5761), confirming that adding icd10_description did not degrade discrimination ability while delivering a meaningful precision gain.

5. Impact of Adding ICD Description

A controlled comparison between Model A (with icd10_description) and Model B (without icd10_description) isolates the precise impact of including the diagnosis label column:

Metric	With ICD Desc (Model A)	Without ICD (Model B)	Difference	Interpretation
Accuracy	63.98%	63.41%	+0.57pp	Marginal improvement

Metric	With ICD Desc (Model A)	Without ICD (Model B)	Difference	Interpretation
Recall	28.13%	30.39%	-2.26pp	Slight decrease
Precision	59.57%	57.14%	+2.43pp	Meaningful improvement
F1 Score	0.3821	0.3968	-0.0147	Marginal decrease
ROC-AUC	0.5760	0.5761	-0.0001	No meaningful difference
False Positives	93	111	-18	Fewer false alarms
True Positives	137	148	-11	Slightly fewer caught
Flagged Claims	1,237	1,356	-119	Tighter, more precise flags

The key insight from this comparison is that `icd10_description` shifts the model toward higher precision at a small cost to recall. In practical terms: the billing team receives 119 fewer flags to review, and 93% of those flags are genuine high-risk claims. The 11 fewer true positives represent claims that would need to be identified through other means (e.g. payer-level audits or manual review of high-value claims).

6. No-Duplicates Model Analysis (Model C)

Model C was trained after removing all 480 duplicate claims from the dataset (5,670 claims remaining). The motivation was that duplicate claims dominated Model A's feature importance — `is_duplicate_claim` had a coefficient of 2.9, dwarfing all other features. By removing duplicates, we forced the model to find the genuine clinical and operational drivers of denial.

Key findings from Model C:

Denial rate dropped to 35.15%: Removing 441 denied duplicate claims lowered the overall denial rate, confirming that duplicates were a disproportionate driver of the 39.58% figure.

Recall improved to 48.12%: Without the dominant duplicate signal, the model distributed its attention more evenly across genuine denial drivers — catching nearly half of all actual non-duplicate denials.

AUC collapsed to 0.4982: The model lost its primary discrimination signal (duplicates) and could not compensate with the remaining, more diffuse features. This confirms that genuine denial prediction without process errors is a genuinely hard problem for Logistic Regression.

New top features emerged: With duplicates removed, the model identified: `department` (Internal Medicine negative, Pediatrics and Emergency positive), `payer_type_Self-Pay`, `is_peak_month`, `provider_denial_rate`, and `days_to_submit` — all actionable operational signals.

2,799 high-risk flags — operationally impractical: Model C flagged 2,799 claims (49% of the dataset) as high-risk. At 35.16% precision, the majority of these flags would be false alarms. A billing team cannot realistically review half of all claims — making this model impractical for day-to-day use without a higher threshold.

7. Feature Importance — Final Model (Model A)

Logistic Regression coefficients reveal both the importance (magnitude) and direction (sign) of each feature's influence on denial probability. Positive coefficients increase denial risk; negative coefficients reduce it.

Feature	Coefficient	Business Interpretation
is_duplicate_claim	+2.900	Highest predictor — duplicates denied at 92%
department_Internal Medicine	-0.161	Lower risk than baseline — well-documented claims
payer_type_Self-Pay	+0.108	Self-Pay payers deny at 40.6% — above average
department_Pediatrics	+0.107	Complex age-specific billing requirements
payer_type_Insurance	-0.091	Insurance payers have established code systems
department_Emergency	+0.081	Unplanned visits lack prior authorisation
is_peak_month	+0.075	March/August high volume increases error rate
provider_denial_rate	+0.068	Historical denial rate is a reliable forward signal
days_to_submit	+0.059	Late submission causes Timely Filing denials
risk_processing	-0.054	Claims with low processing risk are safer

The dominance of `is_duplicate_claim` (coefficient 2.9) is expected and operationally important — duplicate submissions are a fixable process error that accounts for \$1.29M of NovaCare's denial leakage. The remaining top features point to three actionable intervention areas: department-level process improvement (Emergency, Pediatrics), payer relationship management (Self-Pay protocols), and submission timing controls (`days_to_submit`, `is_peak_month`).

8. Model Limitations and Next Steps

Current Limitations

Modest recall (28.13%): The model catches fewer than 3 in 10 actual denials. This is a baseline result — significant improvement is achievable with tree-based models.

Near-random AUC (0.5760): The model's ability to rank claims by denial risk is only marginally better than random. This limits its use as a scoring/ranking tool.

Label encoding for high-cardinality columns: Encoding `provider_name` and `icd10_description` as integers introduces an implicit ordinal relationship that does not exist — a known limitation of label encoding.

Static model: The model was trained on historical data. As NovaCare's payer mix, coding practices, and regulations change, the model will need periodic retraining to remain accurate.

Recommended Next Steps

- Upgrade to Random Forest or XGBoost — tree-based models handle non-linear relationships and feature interactions that Logistic Regression cannot, and typically improve recall by 15–25 percentage points.
- Tune the decision threshold — lowering from 0.5 to 0.35 would catch more denials at the cost of more false positives; the optimal threshold depends on the billing team's review capacity.
- Add `denial_reason` as a training feature for the NEXT model batch — once denial reasons are recorded for the current cohort, they become valid training signal for predicting future claim outcomes.

- Implement monthly retraining — schedule model retraining on each new month's claims to ensure the model adapts to changing patterns.
- Separate models per department — Emergency and Pediatrics have distinct denial patterns; department-specific models may yield significantly higher precision for those high-risk teams.

9. Conclusion

This project delivered NovaCare Insurance's first data-driven denial prediction capability. Starting from 6,150 raw claims with inconsistent provider names, department formats and casing issues, Team 3 of DataVerseAfrica Cohort 4.0 built a complete end-to-end pipeline: SQL cleaning, Python pattern analysis, Logistic Regression modelling, and a four-page interactive Power BI dashboard.

The final model (Model A) assigns every claim a denial probability score. Claims scoring above 0.5 are flagged for pre-submission review. At 59.57% precision, nearly 6 in 10 flagged claims are genuine risks — giving the billing team a reliable, prioritised worklist rather than an overwhelming volume of false alarms.

The \$1.04M in estimated leakage from missed denials (350 False Negatives on the test set) defines the improvement target for Model Version 2. With tree-based algorithms and threshold tuning, Team 3 is confident this baseline can be substantially improved in the next project phase.